Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 3 Ω resistor carries 5 A. Find power using both IV and I^2R, showing the implied voltage.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

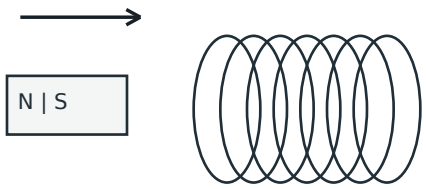
Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

Marzano check - circle: ① ② ③

03 Represent and interpret

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?



Coil

Marzano check - circle: ① ② ③

04 Solve and justify

A 100-turn coil has area .020 m². A uniform perpendicular field changes from .10 T to .30 T in .50 s. Find average emf magnitude.

Marzano check - circle: ① ② ③

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05 Analyze the reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Marzano check - circle: 1 2 3

06 Choose a method

Both charge magnitudes become 4 times as large and separation becomes 6 times as large. Express new force as a fraction of the old.

Marzano check - circle: 1 2 3

07 Exit check

The same flux change occurs in one fourth the time. Compare emf magnitude and name a model assumption.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Flux per turn changes from 0.01 to 0.04 Wb in 0.2 s through 50 turns. Find average emf magnitude and describe Lenz's-law direction qualitatively.

Marzano check - circle: 1 2 3